



EXAM PAPERS PRACTICE

Boost your performance and confidence with these topic-based exam questions

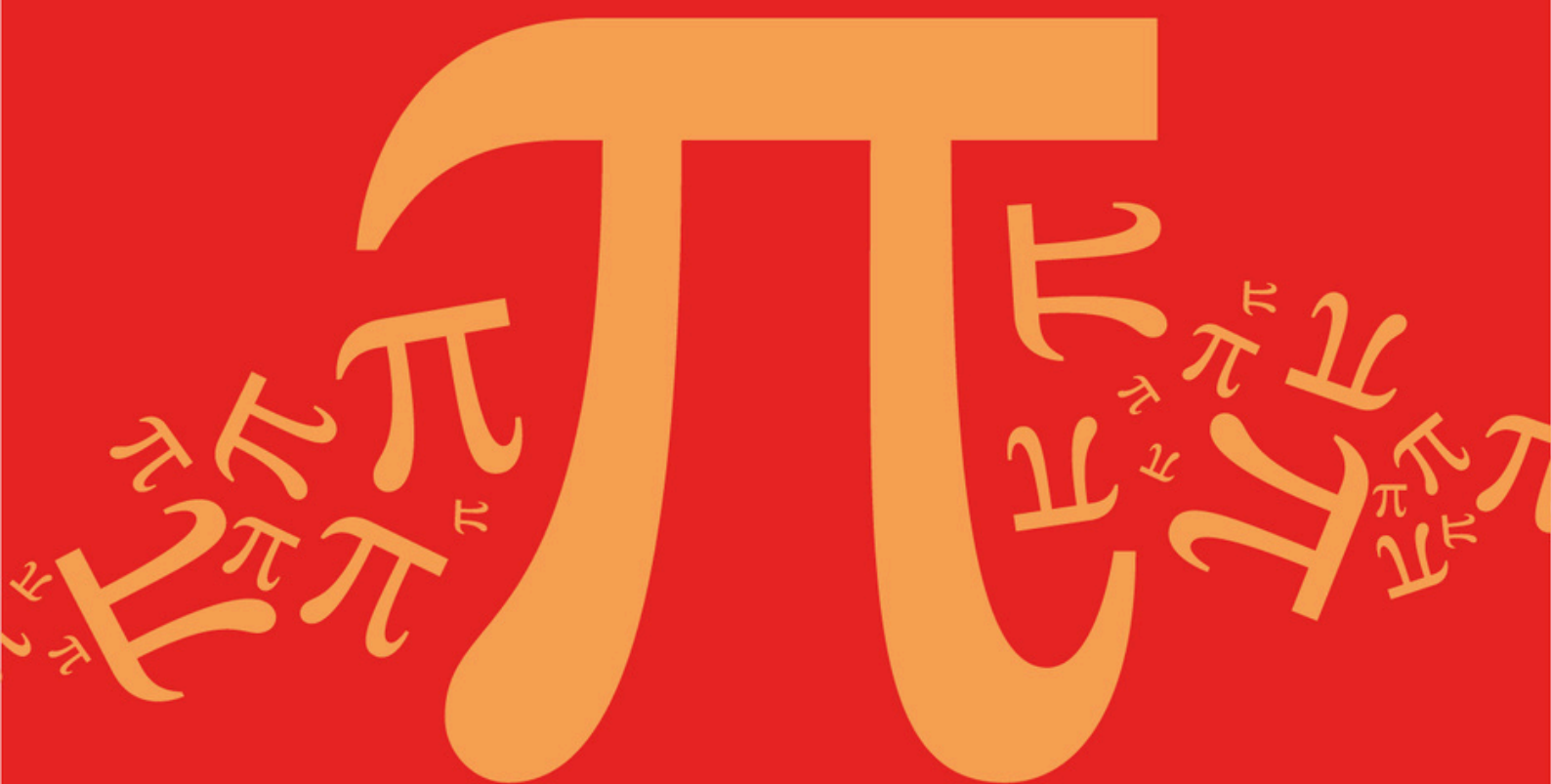
Practice questions created by actual examiners and assessment experts

Detailed mark scheme

Suitable for all boards

Designed to test your ability and thoroughly prepare you

AQA A Level Further Mathematics (7367)



Topic

D: Further Algebra and Functions

Sub topic

D4: Method of Differences

Topic Questions

AQA A Level Further Mathematics (7367)

Question Paper

Topic **D: Further Algebra and Functions**
Sub topic **D4: Method of Differences**

EXAM PAPERS PRACTICE

© 2024 Exams Papers Practice. All Rights Reserved

**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Name : _____

Date : _____

Time : _____

Total Marks Available : _____

Total Marks Received : _____

Q1.

- (a) Show that

$$\frac{1}{r+2} - \frac{1}{r+3} = \frac{1}{(r+2)(r+3)}$$

(1)

- (b) Use the method of differences to show that

$$\sum_{r=1}^n \frac{1}{(r+2)(r+3)} = \frac{n}{3(n+3)}$$

(3)

(Total 4 marks)

Q2.

- (a) Show that $\frac{1}{5r-2} - \frac{1}{5r+3} = \frac{A}{(5r-2)(5r+3)}$, stating the value of the constant A.

(2)

- (b) Hence use the method of differences to show that

$$\sum_{r=1}^n \frac{1}{(5r-2)(5r+3)} = \frac{n}{3(5n+3)}$$

(4)

- (c) Find the value of

$$\sum_{r=1}^{\infty} \frac{1}{(5r-2)(5r+3)}$$

(1)

(Total 7 marks)

Q3.

- (a) Given that $f(r) = r^2(2r^2 - 1)$, show that

$$f(r) - f(r-1) = (2r-1)^3$$

(3)

- (b) Use the method of differences to show that

$$\sum_{r=n+1}^{2n} (2r-1)^3 = 3n^2(10n^2 - 1)$$

(4)

(Total 7 marks)

Q4.

- (a) Show that

$$\frac{2^{r+1}}{r+2} - \frac{2^r}{r+1} = \frac{r2^r}{(r+1)(r+2)}$$

(3)

- (b) Hence find

$$\sum_{r=1}^{30} \frac{r2^r}{(r+1)(r+2)}$$

giving your answer in the form $2^n - 1$, where n is an integer.

(3)

(Total 6 marks)

Q5.

- (a) Given that

$$u_r = \frac{1}{6} r(r+1)(4r+11)$$

show that

$$u_r - u_{r-1} = r(2r+3)$$

(3)

- (b) Hence find the sum of the first hundred terms of the series

$$1 \times 5 + 2 \times 7 + 3 \times 9 + \dots + r(2r+3) + \dots$$

(3)

(Total 6 marks)

Q6.

- (a) Show that

$$(r+1)! - (r-1)! = (r^2 + r - 1 + (r-1)!)!$$

(2)

- (b) Hence show that

$$\sum_{r=1}^n (r^2 + r - 1)(r-1)! = (n+2)n! - 2$$

(4)

(Total 6 marks)

Q7.

- (a) Given that $f(r) = \frac{1}{4}r^2(r+1)^2$, show that

$$f(r) - f(r-1) = r^3$$

(3)

- (b) Use the method of differences to show that

$$\sum_{r=1}^{2n} r^3 = \frac{3}{4}n^2(n+1)(5n+1)$$

(5)

(Total 8 marks)

Q8.

- (a) Show that

$$(2r+1)^3 - (2r-1)^3 = 24r^2 + 2$$

(3)

- (b) Hence, using the method of differences, show that

$$\sum_{r=1}^n r^2 = \frac{1}{6}n(n+1)(2n+1)$$

(6)

(Total 9 marks)

Q9.

- (a) Use the identity $\tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$ with $A = (r+1)x$ and $B = rx$ to show that

$$\tan rx \tan (r+1)x = \frac{\tan (r+1)x}{\tan x} - \frac{\tan rx}{\tan x} - 1$$

(4)

- (b) Use the method of differences to show that

$$\tan \frac{\pi}{50} \tan \frac{2\pi}{50} + \tan \frac{2\pi}{50} \tan \frac{3\pi}{50} + \dots + \tan \frac{19\pi}{50} \tan \frac{20\pi}{50} = \frac{\tan \frac{2\pi}{5}}{\tan \frac{\pi}{50}} - 20$$

(5)

(Total 9 marks)

Q10.

- (a) Given that $f(r) = (r-1)r^2$, show that $f(r+1) - f(r) = r(3r+1)$

(3)

- (b) Use the method of differences to find the value of

$$\sum_{r=50}^{99} r(3r+1)$$

(4)

(Total 7 marks)

Q11.

(a) Show that

$$\frac{1}{r^2} - \frac{1}{(r+1)^2} = \frac{2r+1}{r^2(r+1)^2}$$

(2)

(b) Hence find the sum of the first n terms of the series

$$\frac{3}{1^2 \times 2^2} + \frac{5}{2^2 \times 3^2} + \frac{7}{3^2 \times 4^2} + \dots$$

(4)

(Total 6 marks)